

# *Ingeniería de soluciones con AI*

*Semestre 2026-2*

***Docente:***  
*Francisco Macaya*

*Noticias:* AI no es solo software, sino hardware.

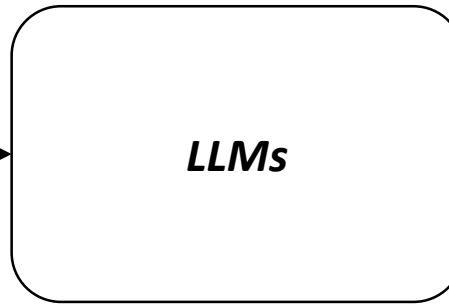


## *Recapitulemos: Los LLMs como un sistema aislado.*

**Prompt Engineering**



*Prompts & Parametros.*

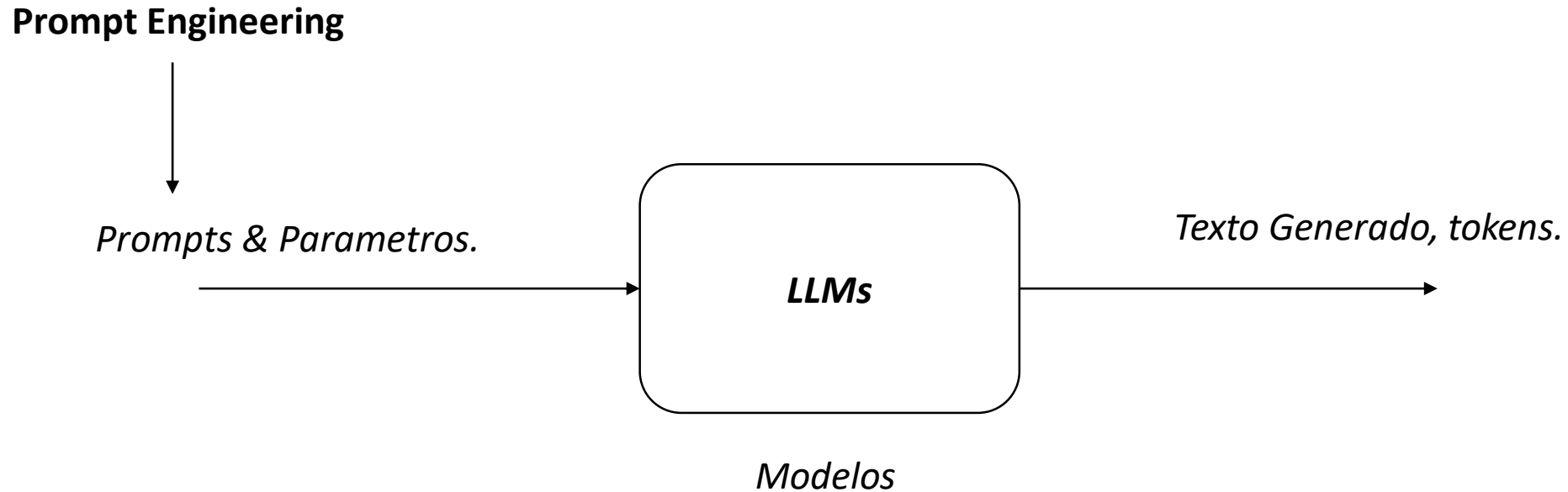


**LLMs**

*Modelos*

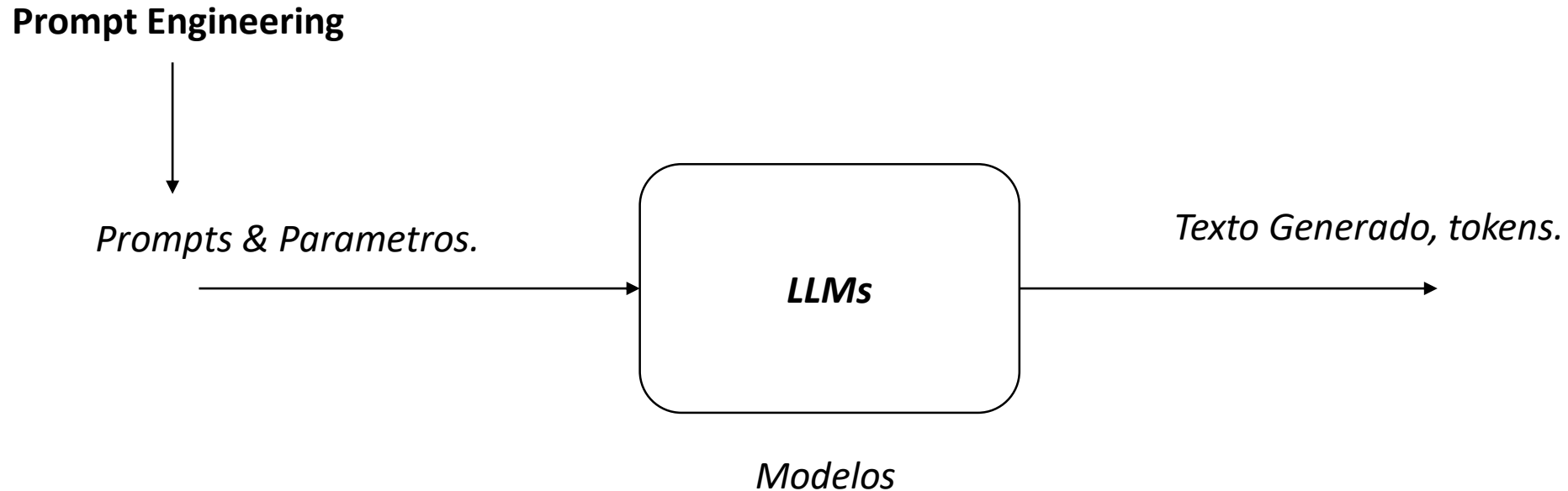
*Texto Generado, tokens.*

## *Recapitulemos: Los LLMs como un sistema aislado.*



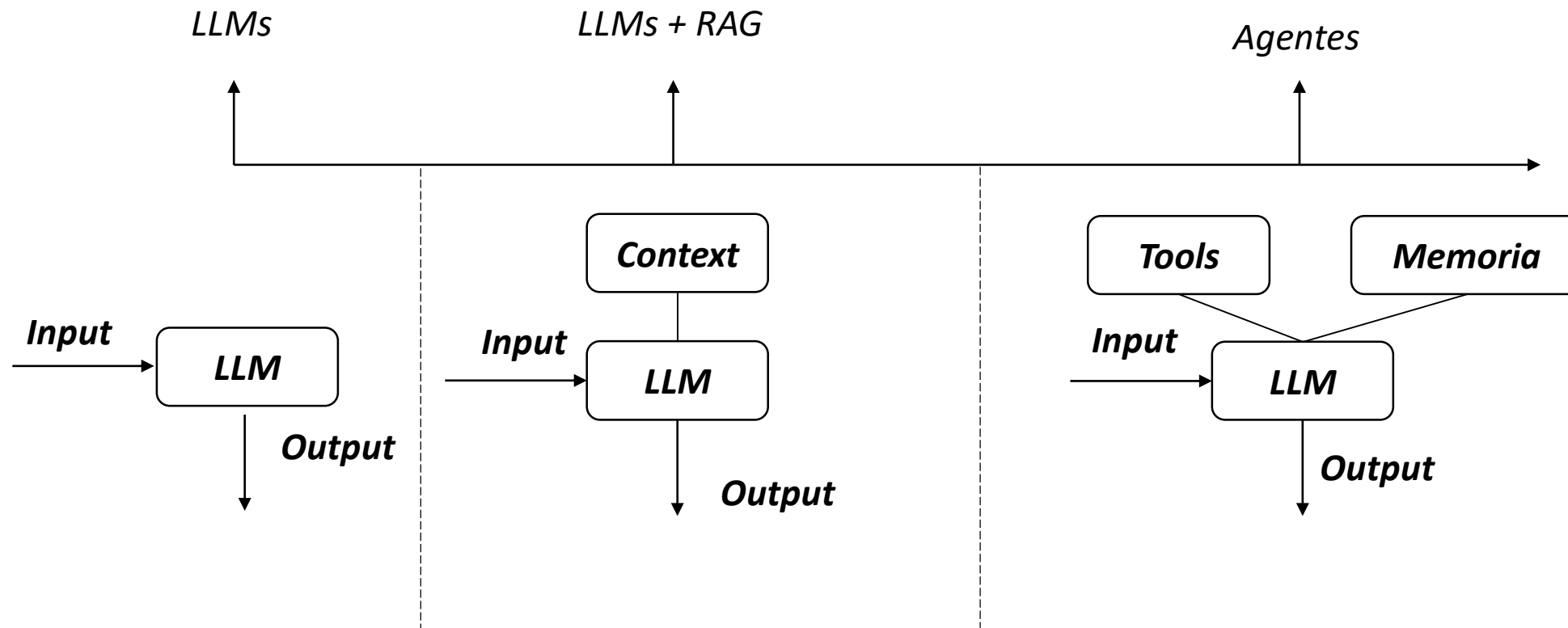
*¿Qué falta?*

## *Recapitulemos: Los LLMs como un sistema aislado.*

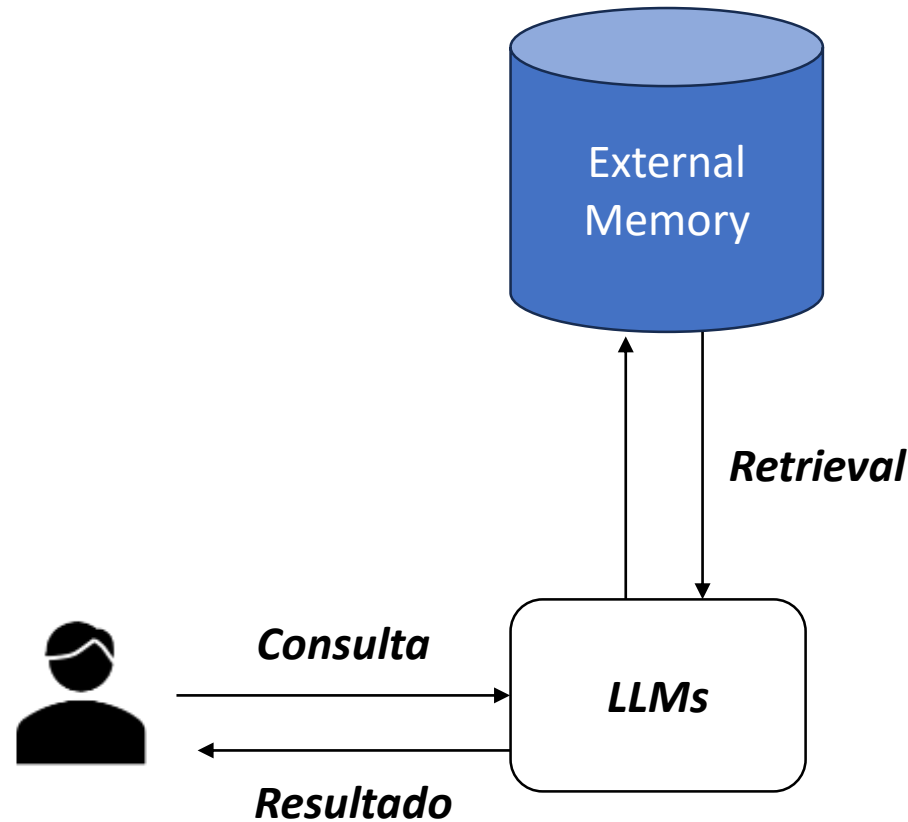


*Los modelos no tienen acceso a datos actuales ni privados. No tienen memoria . No tienen acceso aplicaciones, ni al mundo real. No pueden ejecutar algo. Son como un niño que solo me sirve para tareas generalistas.*

# *De generalista a especialista: El camino de la AI.*



# ¿Qué es RAG?: Retrieval-Augmented Generation



*RAG es una metodología para dotar a los LLMs con información que no tenían al momento de entrenarlos.*

1. **Retrieval (Recuperar):** Encontrar las fuentes relevantes a la pregunta.
2. **Aumentar (Augment):** La información recuperada se inyecta al prompt.
3. **Generar (Generative):** El LLM produce una respuesta en base a la información dada.

# Retrieval: Dos caminos, mismo objetivo.

## ***Term base retrieval***

***Term Frequency (TF)***: Numero de veces que un termino aparece en un documento. Si el termino aparece más veces, es más relevante.

***Inverse document frequency (IDF)*** : La importancia de un termino es inversamente proporcional a la cantidad de documentos donde aparece el termino.



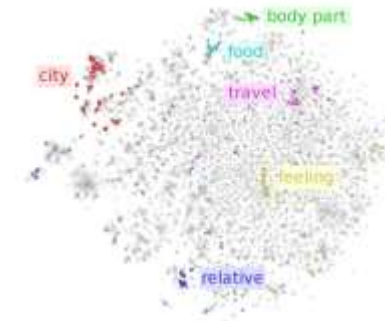
*Las soluciones más típicas a este tipo de retrieval son: **ElasticSearch** y **BM25**.*

***Esta solución se enfoca en el lexical level***

## ***Embedding base retrieval***

Transforma la información en vectores, se encuentran los más parecidos, y luego se inyectan al modelo de lenguaje como parte del prompt.

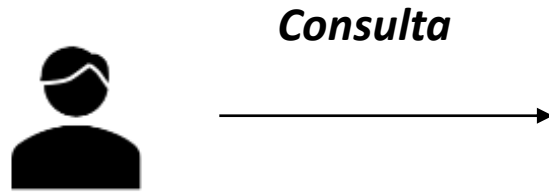
Los vectores son capaces de capturar la representación semántica del texto.



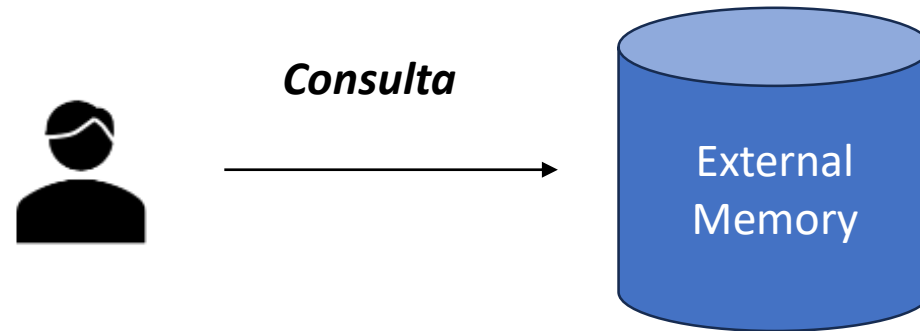
***Esta solución se enfoca en el semantic level.***



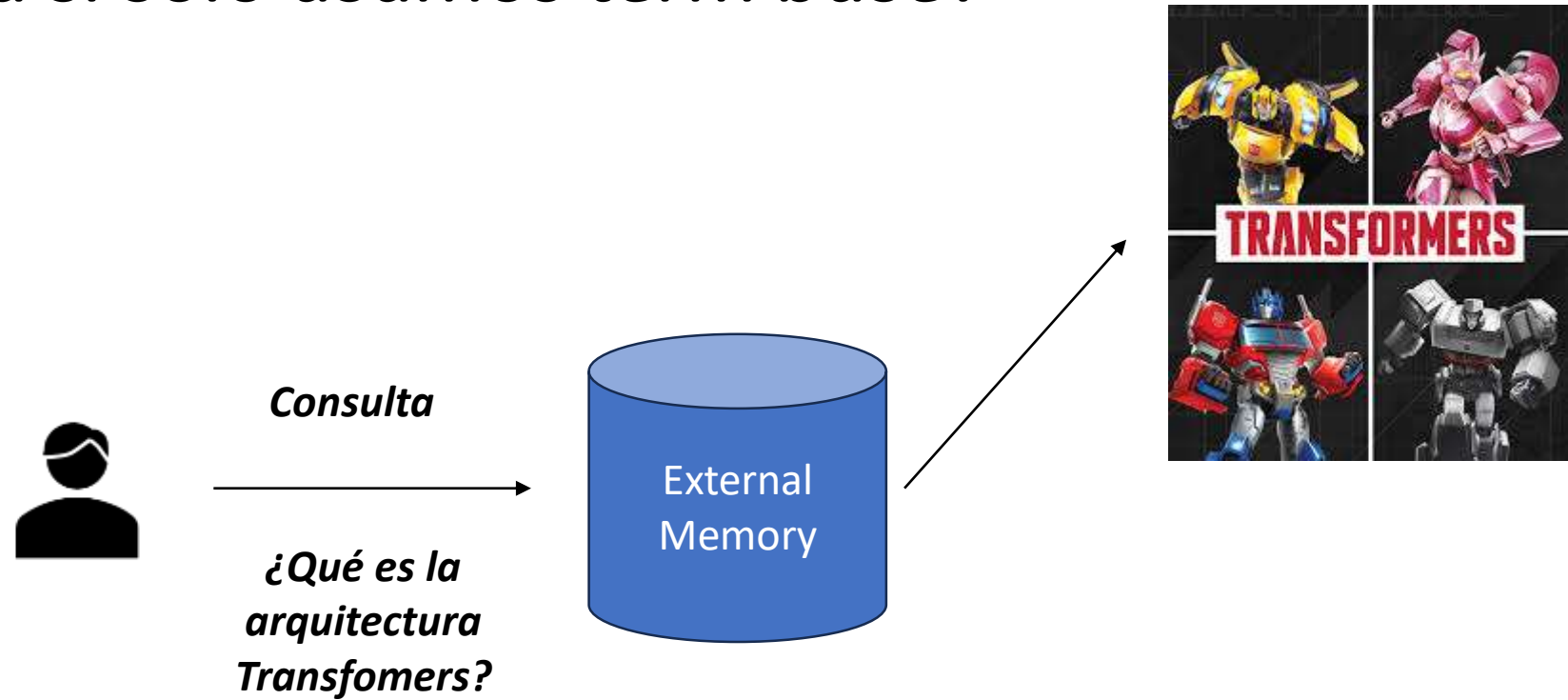
*¿Qué pasaría si solo usamos term base?*



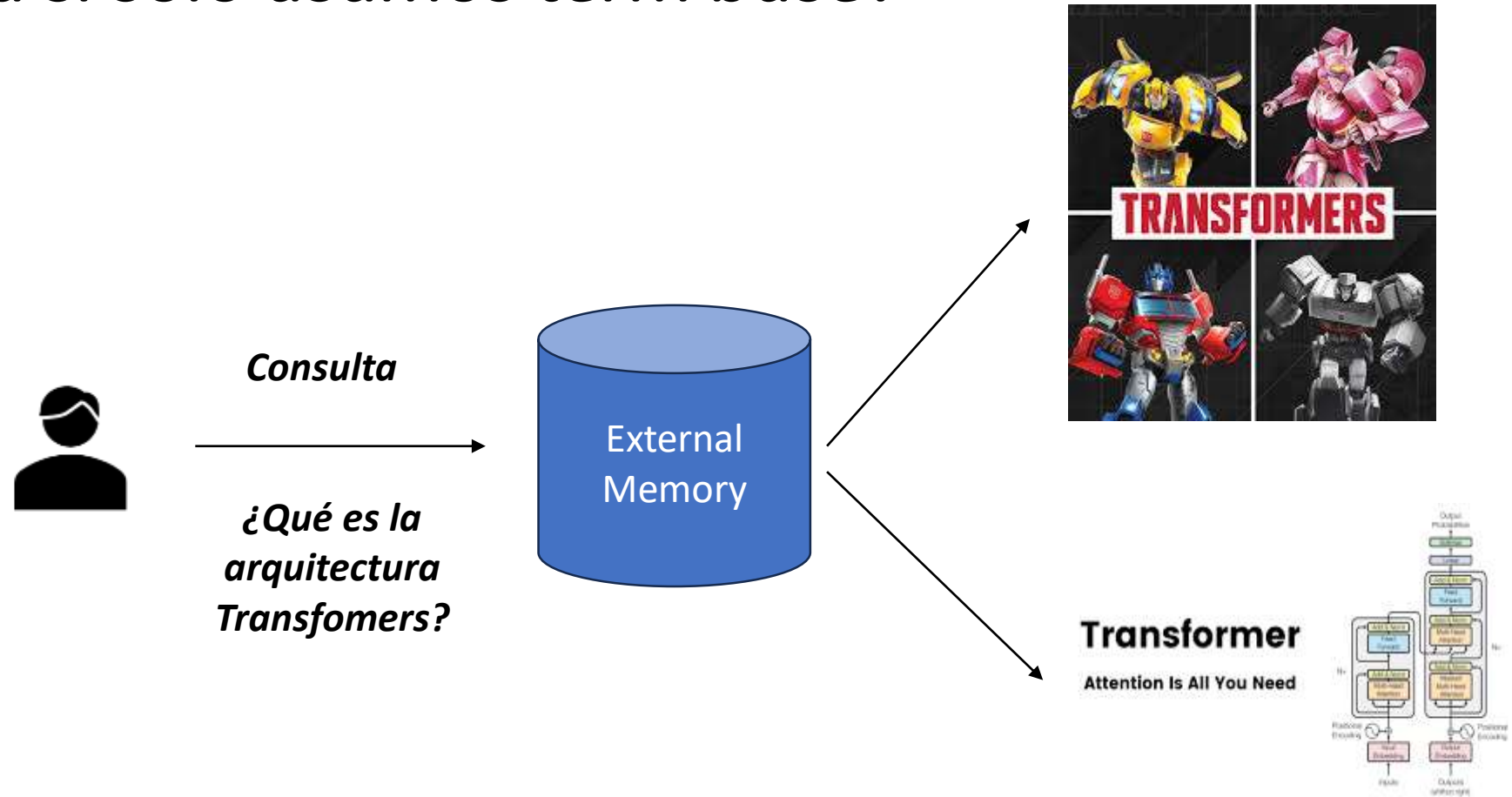
*¿Qué pasaría si solo usamos term base?*



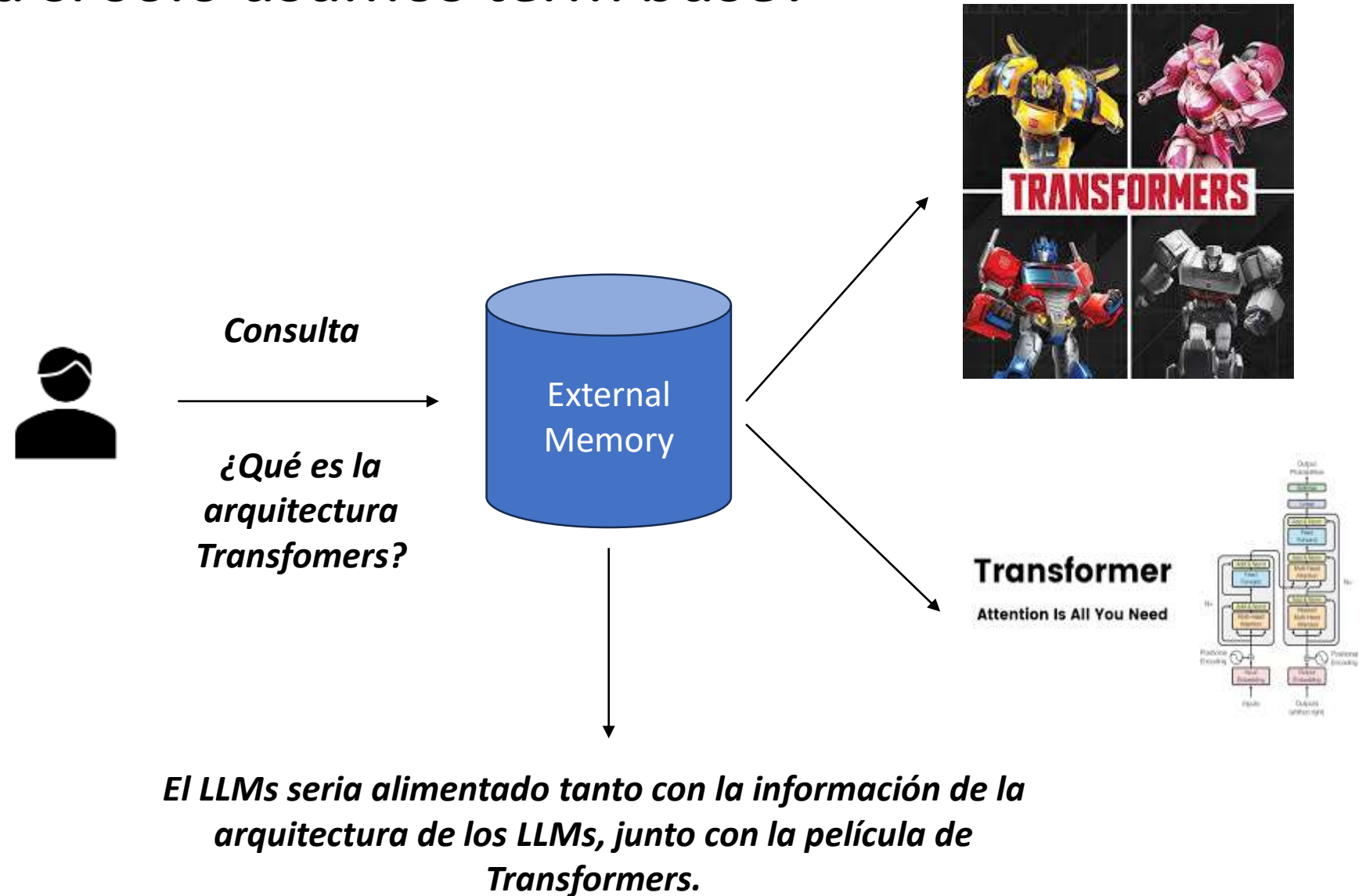
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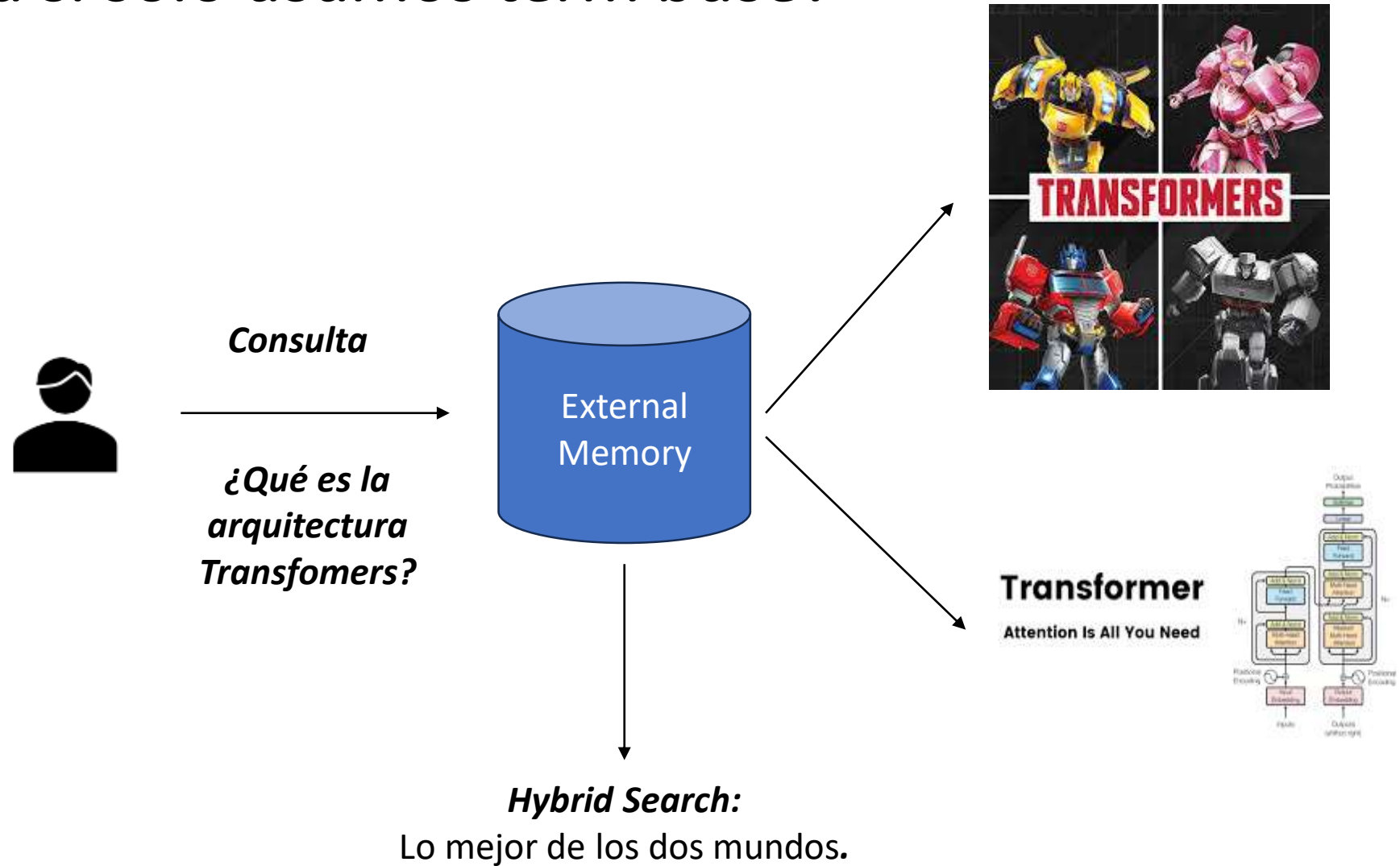
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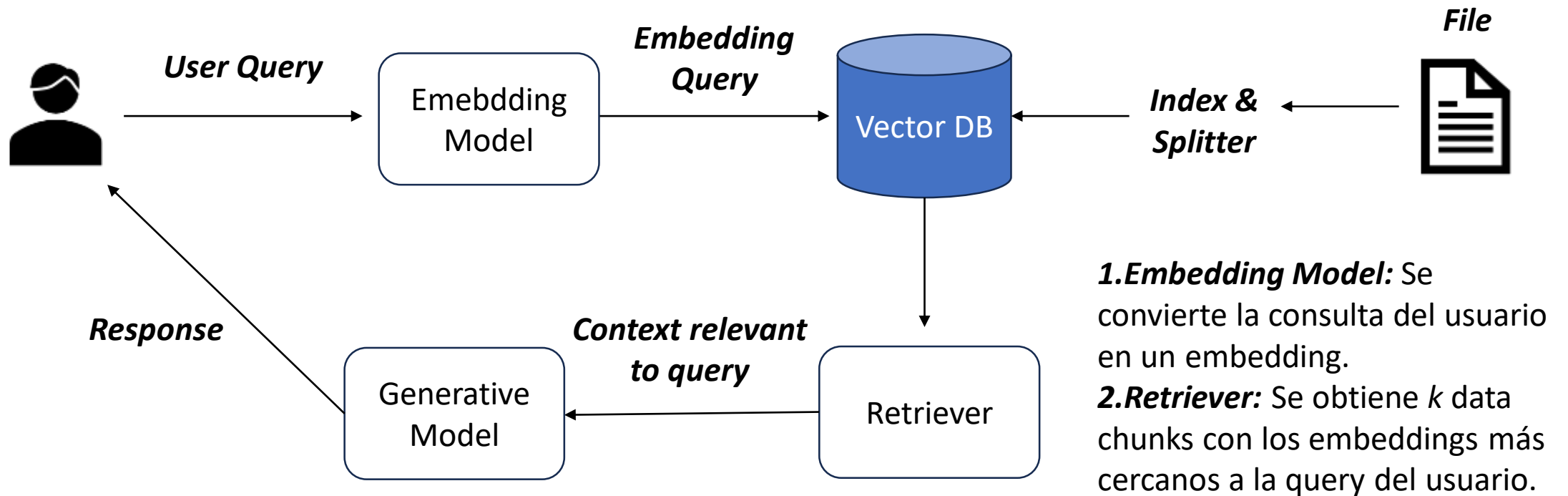
# ¿Qué pasaría si solo usamos term base?



# ¿Qué pasaría si solo usamos term base?



# Arquitectura de RAG:



# Divide and Conquer: Divide el texto en fragmentos.

Los archivos son divididos en **chunks**, lo que permite localizar y recuperar únicamente la información relevante para cada consulta.

**1. Chunk size** — el tamaño de cada fragmento (medido en tokens o caracteres). Define cuánto texto va en cada chunk. Si es muy grande pierdes precisión en el retrieval; si es muy pequeño pierdes contexto.

**2. Chunk overlap** — cuántos tokens/caracteres se repiten entre un chunk y el siguiente. Sirve para evitar que una idea importante quede cortada justo en el borde entre dos chunks.

Large language models (LLMs) are a type of artificial intelligence algorithm that uses deep learning techniques and massive data sets to understand, summarize, generate and predict new content. The "large" in LLM refers to the vast amount of data these models are trained on – often terabytes of text and data – and the number of parameters they use to process language, which can be billions.

LLMs are built on transformer architectures, a neural network architecture introduced in 2017 that processes input data in a way that allows for parallelization, making training on massive data sets feasible. This architecture uses self-attention mechanisms to weigh the importance of different words in the input sequence when processing a specific word.

There are different types of LLMs, categorized based on their architecture and training objectives:

• **"Generative LLMs:"** These models are designed to generate new content, such as text, code, or images. Examples include GPT-3, LLaMA, and Stable Diffusion.

• **"Discriminative LLMs:"** These models are used for classification and analysis tasks, such as sentiment analysis or named entity recognition.

• **"Hybrid LLMs:"** These models combine aspects of both generative and discriminative models.

Chunking is a crucial step in preparing text data for LLMs, especially for tasks like RAG. It involves breaking down a large text document into smaller, manageable pieces or "chunks." The size and method of chunking can significantly impact the performance of LLM applications.



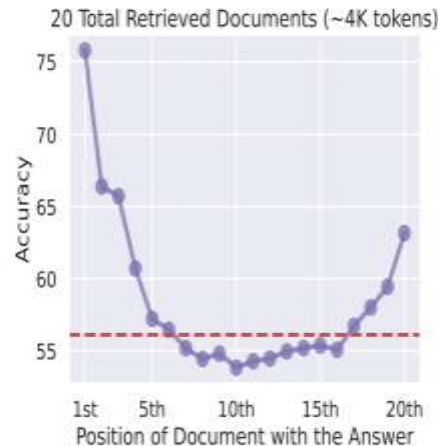
# Los LLMs se pierden con infinito contexto.

## Lost in the Middle: How Language Models Use Long Contexts

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 Michele Bevilacqua<sup>3</sup> Fabio Petroni<sup>3</sup> Percy Liang<sup>1</sup>  
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### Abstract

While recent language models have the ability to take long contexts as input, relatively little is known about how well they use longer context. We analyze the performance of language models on two tasks that require identifying relevant information in their input contexts: multi-document question answering and key-value retrieval. We find that performance can degrade significantly when changing the position of relevant information, indicating that current language models do not robustly make use of information in



length of 4K tokens, and GPT-3.5-Turbo (16K) is a version with an extended maximum context length of 16K tokens. We evaluate Claude-1.3 and Claude-1.3 (100K) with the Anthropic API; Claude-1.3 has a maximum context length of 8K tokens, and Claude-1.3 (100K) has an extended context length of 100K tokens.<sup>6</sup>

### 2.3 Results and Discussion

We experiment with input contexts containing 10, 20, and 30 total documents. Figure 5 presents multi-document question answering performance when varying the position of relevant information within the input context. To contextualize model performance, we also evaluate on the closed-book and oracle settings (Table 1). In the closed-book setting, models are not given any documents in their input context, and must rely on their parametric memory to generate the correct answer. On the other hand, in the oracle setting, language models are given the single document that contains the answer and must use it to answer the question.

**Model performance is highest when relevant information occurs at the beginning or end of its input context.** As illustrated in Figure 5, changing the position of relevant information in the input context leads to substantial decreases in model performance. In particular, we see a distinctive U-

<sup>5</sup>We use the 0613 OpenAI model versions.

<sup>6</sup>We also evaluate GPT-4 (8K) on a subset of multi-document QA experiments, finding similar results and trends as other models (though GPT-4 has higher absolute performance). Evaluating GPT-4 on the full multi-document QA and key-value retrieval experiments would cost upwards of \$6000. See Appendix D for GPT-4 results and discussion.

|                     |       |       |
|---------------------|-------|-------|
| LongChat-13B (16K)  | 35.0% | 83.4% |
| MPT-30B-Instruct    | 31.5% | 81.9% |
| GPT-3.5-Turbo       | 56.1% | 88.3% |
| GPT-3.5-Turbo (16K) | 56.0% | 88.6% |
| Claude-1.3          | 48.3% | 76.1% |
| Claude-1.3 (100K)   | 48.2% | 76.4% |

Table 1: Closed-book and oracle accuracy of language models on the multi-document question answering task.

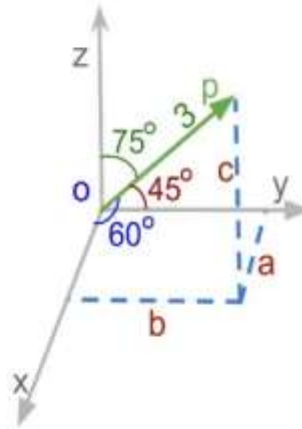
shaped performance curve—models are often much better at using relevant information that occurs at the very beginning (primacy bias) and very end of contexts (recency bias), and suffer degraded performance when forced to use information within the middle of its input context. For example, GPT-3.5-Turbo’s multi-document QA performance can drop by more than 20%—in the worst case, performance in 20- and 30-document settings is lower than performance without *any* input documents (i.e., closed-book performance; 56.1%). These results indicate that current models cannot effectively reason over their entire context window when prompted for downstream tasks.

**Extended-context models are not necessarily better at using input context.** When the input context fits in the context window of both a model and its extended-context counterpart, we see that performance between them is nearly identical. For example, the 10- and 20-document settings both fit in the context window of GPT-3.5-Turbo and GPT-3.5-Turbo (16K), and we observe that their performance as a function of position of relative information is nearly superimposed (solid purple and dashed brown series in Figure 5). These results

# Similitud de Coseno: “Cercanía” en los chunks.



$$\text{similitud}(\vec{a}, \vec{b}) = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|}$$



La similitud de coseno es la manera para determinar algebraicamente si dos vectores son similares o no.

El resultado va de **-1 a 1**:

- **1** = vectores idénticos (misma dirección)
- **0** = vectores perpendiculares (sin relación)
- **-1** = vectores opuestos

Los embeddings de texto producen vectores cuyos valores de similitud coseno van de 0 a 1 en la práctica, donde valores cercanos a 1 indican mayor similitud semántica entre los textos comparados.

*Bases vectoriales: Se han popularizado muchas en el ultimo tiempo.*



**Chroma**



**Pinecone**



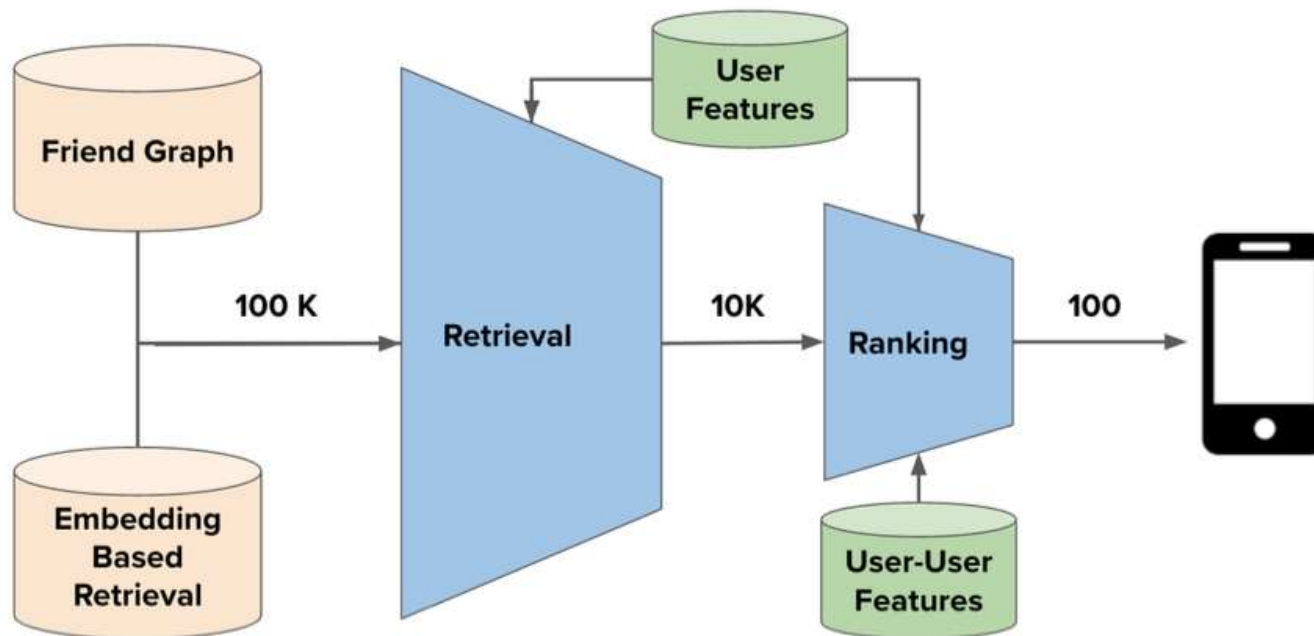
**Weaviate**

**FAISS**  
(FACEBOOK AI SIMILARITY SEARCH)



**qdrant**

# Reranking: El “reordenamiento” de los chunks



*Los sistemas de recomendación funcionan en dos etapas: retrieval y el ranking.*

*En base a esto, la distancia de coseno era la retrieval, ahora faltaba la parte de ranking.*

*Los investigadores y laboratorios encontraron que la búsqueda vectorial no era lo suficiente, por lo que, plantean añadirle otros concepto, que son los **modelos de reranking**.*



# Reranking: El “reordenamiento” de los chunks

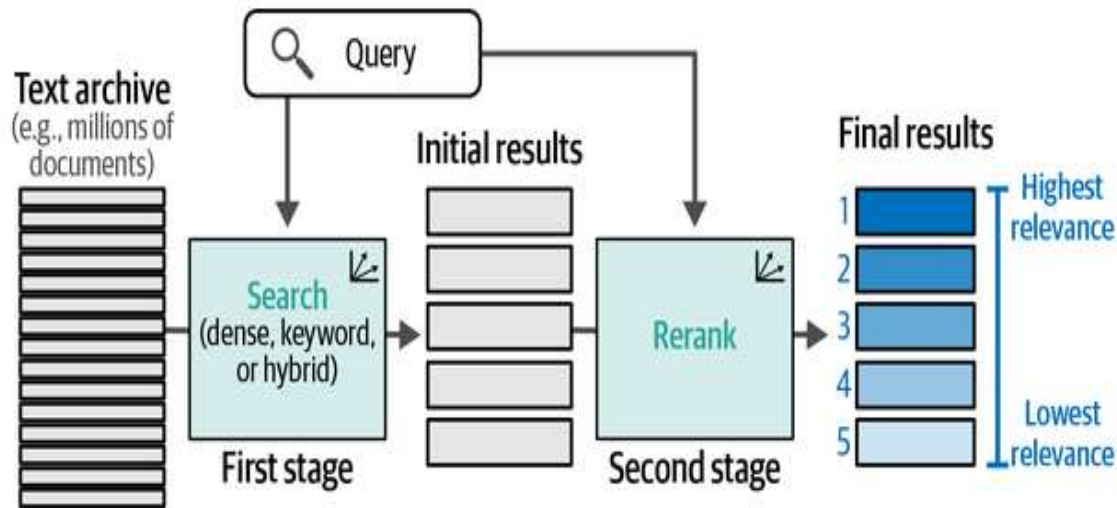


Figure 8-14. LLM rerankers operate as part of a search pipeline with the goal of reordering a number of shortlisted search results by relevance.

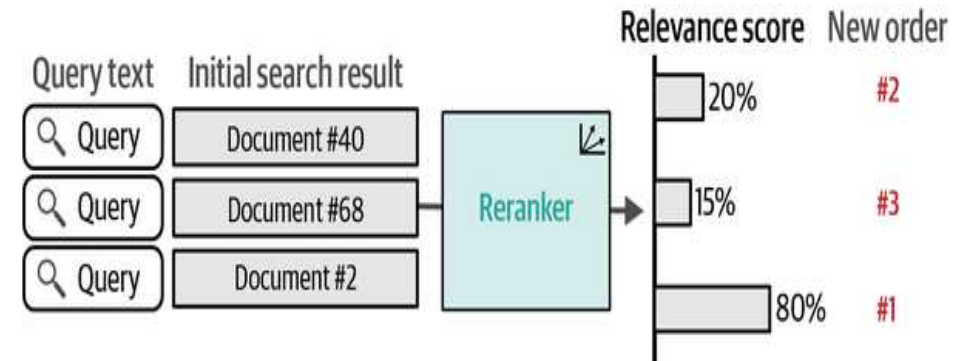


Figure 8-15. A reranker assigns a relevance score to each document by looking at the document and the query at the same time.

*La consulta es presentada a un modelo de LLM simultáneamente con los otros documentos, asignado un score de relevancia por cada fragmento.*

# ¿How it Works?: Documentos y consulta.

```

1 def keyword_and_reranking_search(query, top_k=3, num_candidates=10):
2     print("Input question:", query)
3
4     ##### BM25 search (lexical search) #####
5     bm25_scores = bm25.get_scores(bm25_tokenizer(query))
6     top_n = np.argpartition(bm25_scores, -num_candidates)[-num_candidates:]
7     bm25_hits = [{ 'corpus_id': idx, 'score': bm25_scores[idx]} for idx in top_n]
8     bm25_hits = sorted(bm25_hits, key=lambda x: x['score'], reverse=True)
9
10
11     print(f"Top-{top_k} lexical search (BM25) hits")
12     for hit in bm25_hits[:top_k]:
13         print("\t{:.3f}\t{}".format(hit['score'], texts[hit['corpus_id']].replace("\n", " ")))
14
15
16     ##### Re-ranking #####
17     docs = [texts[hit['corpus_id']] for hit in bm25_hits]
18
19     print(f"\nTop-{top_k} hits by rank-API ({len(bm25_hits)} BM25 hits re-ranked)")
20     results = co.rerank(query=query, documents=docs, top_n=top_k, return_documents=True)
21
22     for hit in results.results:
23         print("\t{:.3f}\t{}".format(hit.relevance_score, hit.document.text.replace("\n", " ")))
24
25
26
27
28
29
30
31

```

```
keyword_and_reranking_search(query = "how precise was the science")
```

*Una consulta, un conjunto de documentos y un score de relevancia para los chunks*

*Reranking: La industria a popularizado empresas especializadas en modelos de reranking.*



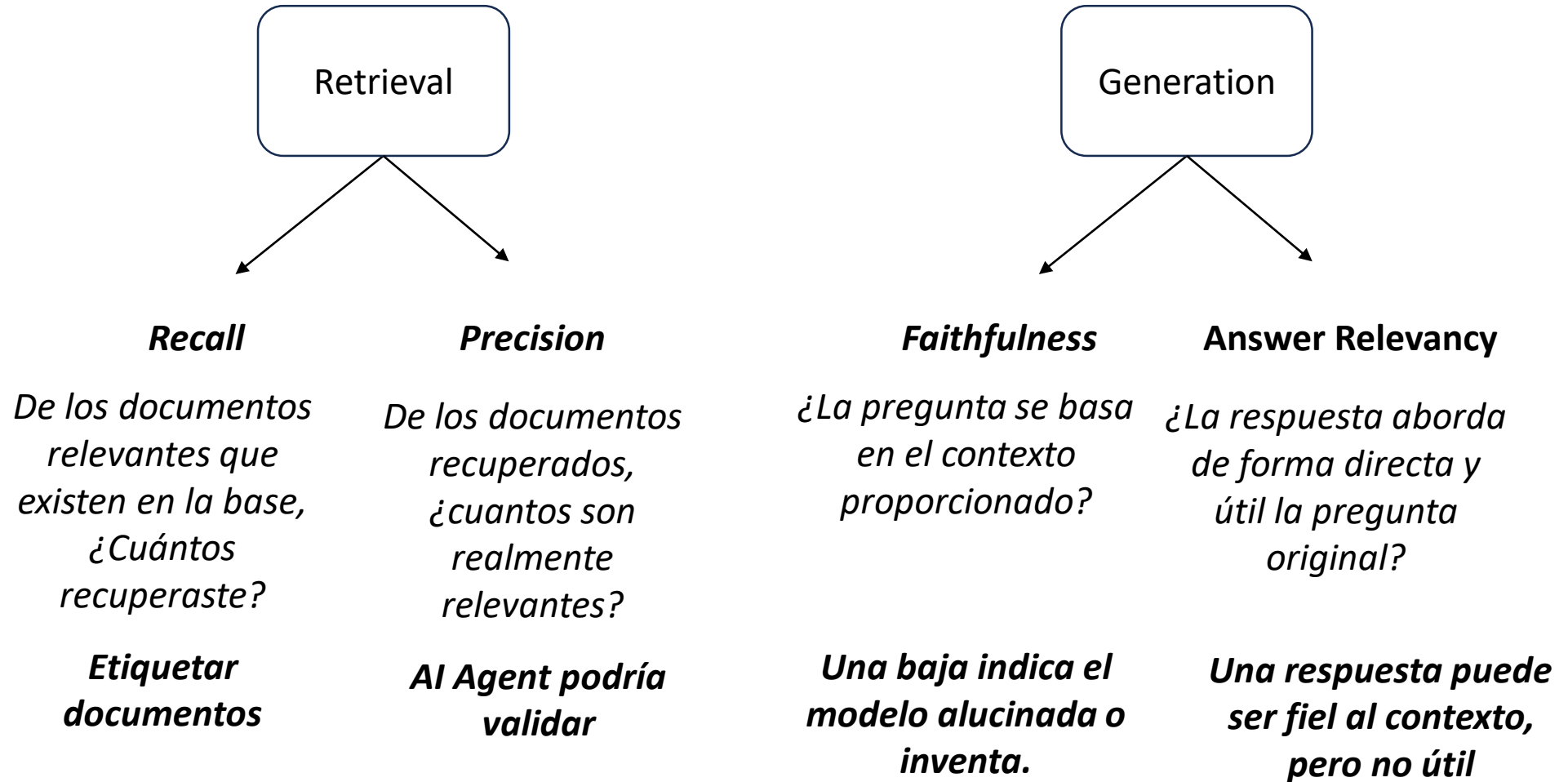
*¿Qué falta aquí?*



*¿Qué falta aquí?*

***Lo que se mide, se gestiona. Peter Drucker (1909-2005)***

# *Control: Métricas de Recuperación y Generación.*



# RAGAS: Librería especializada para medir RAGs

Ragas Office Hours - If you need help setting up Evals for your AI application, sign up for our Office Hours [here](#).

 Ragas

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 vibrantlabsai/ragas  
☆ 13k 📄 1.3k

 Get Started  Core Concepts  How-to Guides  References  Community

## Introduction

Copy page

Ragas is a library that helps you move from "vibe checks" to systematic evaluation loops for your AI applications. It provides tools to supercharge the evaluation of Large Language Model (LLM) applications, enabling you to evaluate your LLM applications with ease and confidence.

### Why Ragas?

Traditional evaluation metrics don't capture what matters for LLM applications. Manual evaluation doesn't scale. Ragas solves this by combining **LLM-driven metrics** with **systematic experimentation** to create a continuous improvement loop.

### Key Features

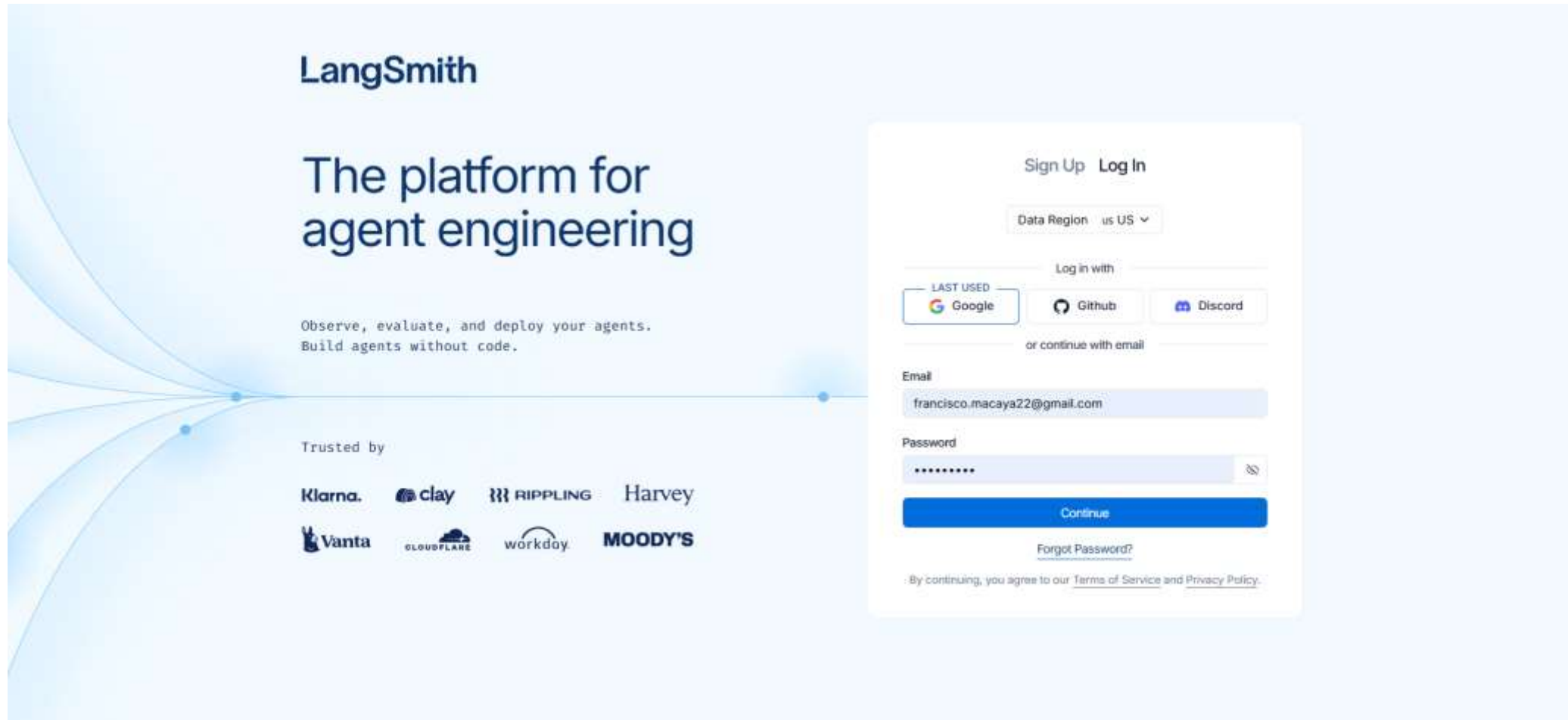
- **Experiments-first approach:** Evaluate changes consistently with `experiments`. Make changes, run evaluations, observe results, and iterate to improve your LLM application.
- **Ragas Metrics:** Create custom metrics tailored to your specific use case with simple decorators or use our library of [available metrics](#). Learn more about [metrics in Ragas](#).
- **Easy to integrate:** Built-in dataset management, result tracking, and integration with popular frameworks like LangChain, LlamaIndex, and more.

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- Why Ragas?
- Key Features
- Want help improving your AI application using evals?

*Todo parece bien, pero ... como mido realmente.  
Necesito registros de buggs, errores, llamadas a las  
Tools ... etc.*

# LangSmith: Todo en un lugar.



The image shows the LangSmith web interface. On the left, a large blue graphic with curved lines and dots is next to the LangSmith logo and the text 'The platform for agent engineering'. Below this, it says 'Observe, evaluate, and deploy your agents. Build agents without code.' and 'Trusted by' followed by logos for Klarna, clay, RIPPLING, Harvey, Vanta, CLOUDFLARE, workday, and MOODY'S. On the right, a white login/sign-up box contains 'Sign Up' and 'Log In' links, a 'Data Region' dropdown set to 'us US', and a 'Log in with' section with buttons for Google (marked 'LAST USED'), Github, and Discord. Below these are links for 'or continue with email', an email input field with 'francisco.macaya22@gmail.com', a password input field with masked characters, a blue 'Continue' button, a 'Forgot Password?' link, and a footer line: 'By continuing, you agree to our Terms of Service and Privacy Policy.'

LangSmith

The platform for agent engineering

Observe, evaluate, and deploy your agents.  
Build agents without code.

Trusted by

Klarna. clay RIPPLING Harvey  
Vanta CLOUDFLARE workday MOODY'S

Sign Up Log In

Data Region us US

Log in with

LAST USED  
Google Github Discord

or continue with email

Email  
francisco.macaya22@gmail.com

Password  
.....

Continue

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